Tutorial for LLVM Intermediate Representation

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Motivation for Learning LLVM Low-level Language (i.e., Handling Intermediate Representation)

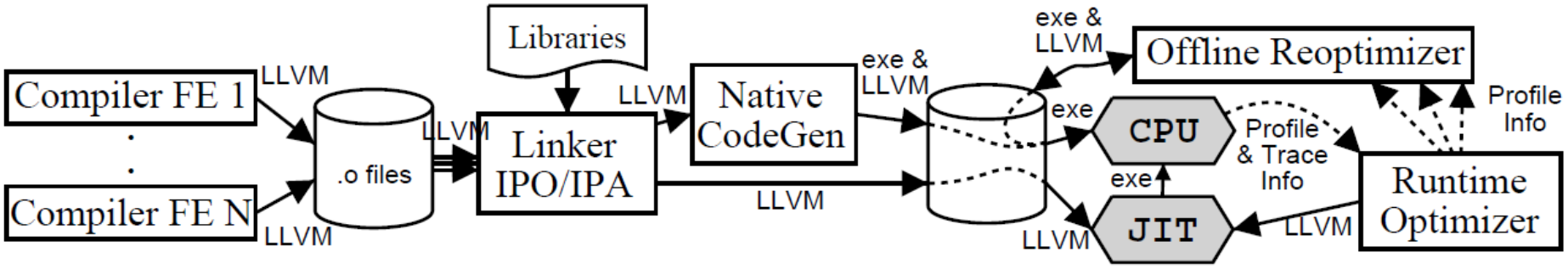
- Biologists know how to **analyze** laboratory mice. In addition, they know how to **modify** the mice by applying new medicine or artificial organ
- Mechanical engineers know how to analyze and modify mechanical products using CAD **tools**.
- Software engineers also have to know how to analyze and modify software code which is far more complex than any engineering product. Thus, software analysis/modification requires **automated analysis tools.**
  - Using source level analysis framework (e.g., Clang, C Intermediate Language (CIL), EDG parser)
  - Using low-level intermediate representation (IR) analysis framework (e.g., LLVM IR)

LLVM is Professional Compiler

- Clang, the LLVM C/C++ front-end supports the full-features of C/C++ and compatible with GCC
- The executable compiled by Clang/LLVM is as fast as the executable by GCC

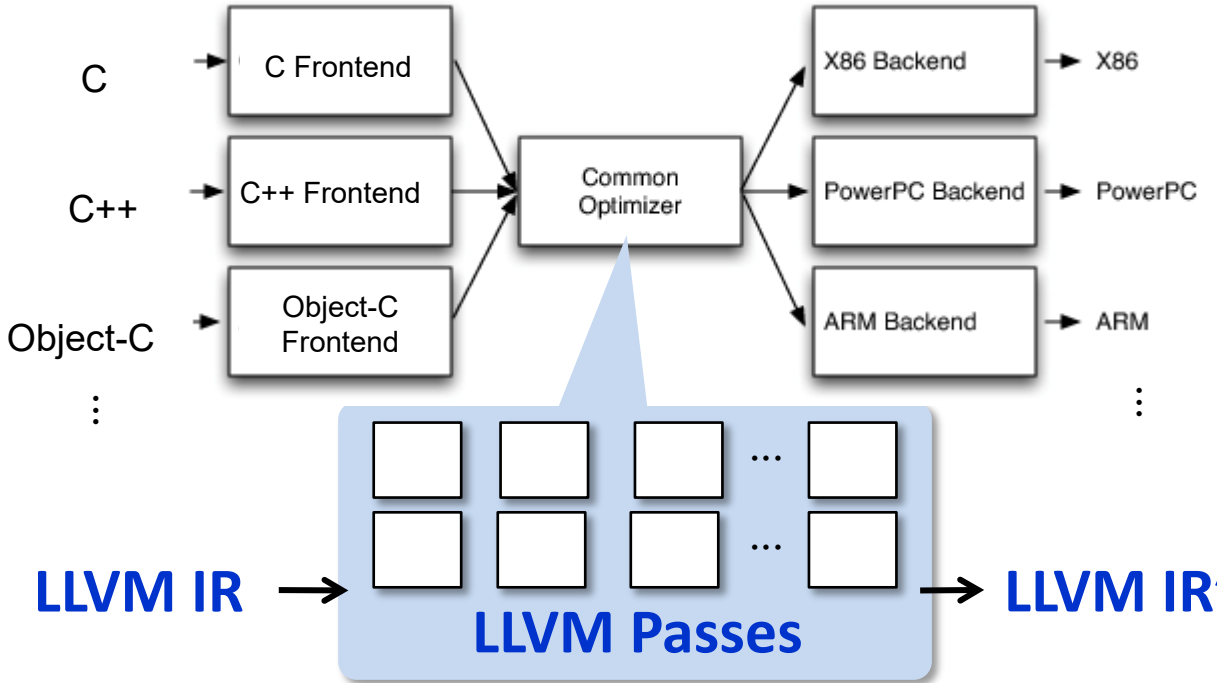
LLVM Compiler Infrastructure (1/2)

- Clang, the LLVM C/C++ front-end supports the full-features of C/C++ and compatible with GCC
- The executable compiled by Clang/LLVM is as fast as the executable by GCC



LLVM Compiler Infrastructure (2/2)

- A collection of modular compilers and analyzers written in C++ with STL.
- LLVM provides hundreds Passes http://llvm.org/docs/Passes.html
  - Analyzers: alias analysis, call graph constructions, dependence analysis, etc.
  - Transformers : dead code elimination, function inlining, constant propagation, loop unrolling, etc.
  - Utilities: CFG viewer, basic block extractor, etc.



LLVM IR as Analysis Target

- The LLVM IR of a program is a *better target for analysis and engineering* than the program source code.
  - Language-independent
    - Able to represent C/C++/Object-C programs
  - Simple
    - register machine
      - Infinite set of typed virtual registers
      - 3-address form instruction
      - Only 31 instruction opcodes
    - static single assignment (SSA)
    - composed as basic blocks
  - Informative
    - typed language
    - control-flow
- LLVM IR is also called as LLVM language, assembly, bitcode, bytecode, code representation

LLVM IR At a Glance

C program language	*LLVM IR*
• Scope: *file, function*	*module, function*
• Type: *bool, char, int, struct{int, char}*	*i1, i8, i32, {i32, i8}*
• A statement with multiple expressions	A sequence of instructions each of which is in a form of "x = y *op* z".
• Data-flow: a sequence of reads/writes on variables	1. load the values of memory addresses (variables) to registers; 2. compute the values in registers; 3. store the values of registers to memory addresses * each register must be assigned exactly once (SSA)
• Control-flow in a function: if, for, while, do while, switch-case,…	A set of basic blocks each of which ends with a conditional jump (or return)

Example

simple.c

```
1  #include <stdio.h>
2  int x, y ;
3
4  int main() {
5    int t ;
6    scanf("%d %d",&x,&y);
7    t = x - y ;
8    if (t > 0)
9      printf("x > y") ;
10   return 0 ;
11 }
```

```
$ clang -S -emit-llvm
-fno-discard-value-names simple.c
```

simple.ll (simplified)

```
…
2   6 @x = common global i32 0, align 4
    7 @y = common global i32 0, align 4

4   11 define i32 @main() #0 {
    12 entry:

5   …
    14 %t = alloca i32, align 4

6   …
    16 %call = call i32 (ptr, ...)
         @__isoc99_scanf(…ptr @x,ptr @y)

7   17 %0 = load ptr @x, align 4
    18 %1 = load ptr @y, align 4
    19 %sub = sub nsw i32 %0 %1
    20 store i32 %sub, ptr %t, align 4

8   21 %2 = load i32, ptr %t, align 4
    22 %cmp = icmp sgt i32 %2, 0
    23 br i1 %cmp, label %if.then,
                   label %if.end

9   24 if.then:
    25   %call1 = call i32 … @printf(…
    26   br label %if.end

10  27 if.end:
    28   ret i32 0
```

Contents

- LLVM IR Instruction
  - architecture, static single assignment
- Data representation (page 13)
  - types, constants, registers, variables
  - load/store instructions, cast instructions
  - computational instructions
- Control representation (page 32)
  - control flow (basic block)
  - control instructions

* *LLVM Language Reference Manual* http://llvm.org/docs/LangRef.html
* *Mapping High-Level Constructs to LLVM IR*
  http://llvm.lyngvig.org/Articles/Mapping-High-Level-Constructs-to-LLVM-IR

LLVM IR Architecture

- RISC-like instruction set
  - Only 31 op-codes (types of instructions) exist
  - Most instructions (e.g. computational instructions) are in three-address form: one or two operands, and one result
- Load/store architecture
  - Memory can be accessed via load/store instruction (for variables)
  - Computational instructions operate on registers (for values)
- Infinite and typed *virtual registers*
  - It is possible to declare a new register any point (the backend maps virtual registers to physical ones).
  - A register is declared with a primitive type (boolean, int, float, pointer)

Static Single Assignment (1/2)

- In SSA, each variable is assigned exactly once, and every variable is defined before its uses.
- Conversion
  - For each definition, create a new version of the target variable (left-hand side) and replace the target variable with the new variable.
  - For each use, replace the original referred variable with the versioned variable reaching the use point.

```
1   x = y + x ;
2   y = x + y ;
3   if (y > 0)
4     x = y ;
5   else
6     x = y + 1 ;
```



```
11  x1 = y0 + x0 ;
12  y1 = x1 + y0 ;
13  if (y1 > 0)
14    x2 = y1 ;
15  else
16    x3 = y1 + 1 ;
```

Static Single Assignment (2/2)

- Use ϕ function if two versions of a variable are reaching one use point at a joining basic block
  - $\phi(x_1, x_2)$ returns a either x_1 or x_2 depending on which block was executed

```
1   x = y + x ;
2   y = x + y ;
3   if (y > 0)
4     x = y ;
5   else
6     x = y + 1 ;
7   y = x - y ;
```



```
11 x1 = y0 + x0 ;
12 y1 = x1 + y0 ;
13 if (y1 > 0)
14   x2 = y1 ;
15 else
16   x3 = y1 + 1 ;
17 x4 = φ(x2, x3);
18 y2 = x4 - y1 ;
```

Data Representations

- Primitive types
- Constants
- Registers (virtual registers)
- Variables
  - local variables, heap variables, global variables
- Load and store instructions
- Aggregated types

Primitive Types

- Language independent primitive types with predefined sizes
  - void: **`void`**
  - bool: **`i1`**
  - integers: **`i[N]`** where $\mathbb{N}$ is 1 to $2^{23}-1$
    e.g. **`i8`**, **`i16`**, **`i32`**, **`i1942`**
  - floating-point types:
    `half` (16-bit floating point value)
    `float` (32-bit floating point value)
    `double` (64-bit floating point value)

- All pointers have one identical **`ptr`** type

Constants

- Boolean (`i1`): **`true`** and **`false`**
- Integer: standard integers including negative numbers
- Floating point: decimal notation, exponential notation, or hexadecimal notation (IEEE754 Std.)
- Pointer: **`null`** is treated as a special value

Registers

- Identifier syntax
  - Named registers: `[%][a-zA-Z$._][a-zA-Z$._0-9]*`
  - Unnamed registers: `[%][0-9][0-9]*`
- A register has a function-level scope.
  - Two registers in different functions may have the same identifier
- A register is assigned for a particular type and a value at its first (and the only) definition

Variables

- In LLVM, all addressable objects (“lvalues”) are explicitly allocated.
- Global variables
  - Each variable has a global scope symbol that points to the memory address of the object
  - Variable identifier: **`[@][a-zA-Z$._][a-zA-Z$._0-9]*`**
- Local variables
  - The **`alloca`** instruction allocates memory in the stack frame.
  - Deallocated automatically if the function returns.
- Heap variables
  - The **`malloc`** function call allocates memory on the heap.
  - The **`free`** function call frees the memory allocated by **`malloc`**.

Load and Store Instructions

- Load

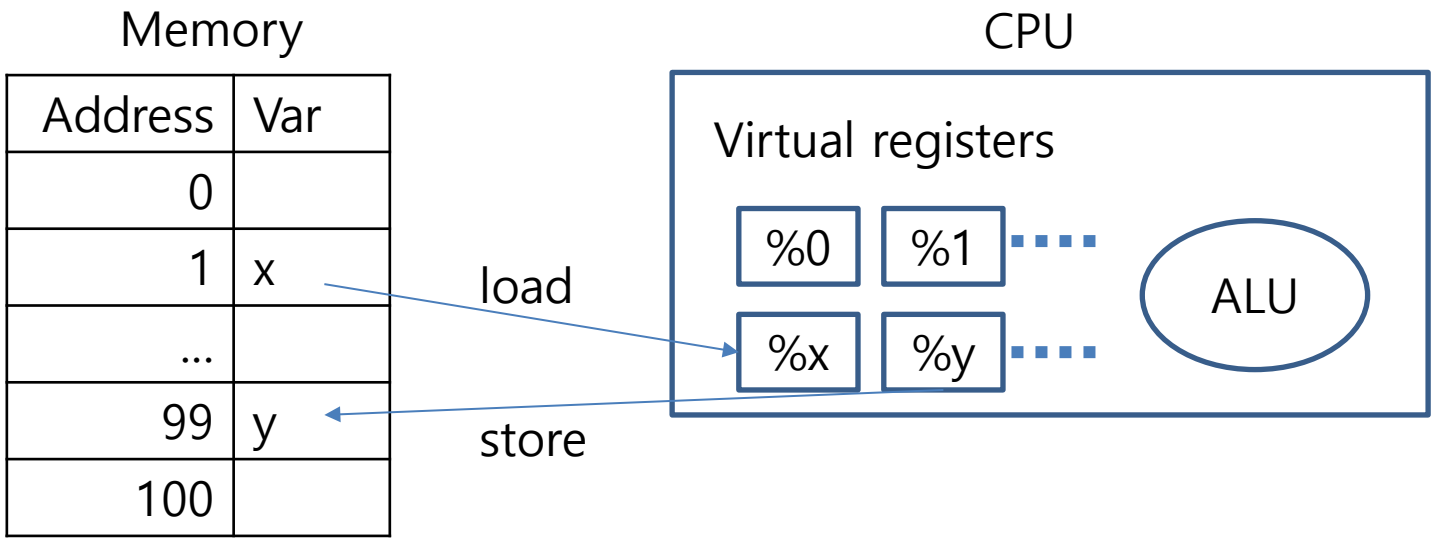
```
<result>=load <type>, ptr <ptr>
```

  - result: the target register
  - type: the type of the data (a pointee value type)
  - ptr: the register that has the address of the data

- Store

```
store <type> <value>, ptr <ptr>
```

  - type: the type of the value
  - value: either a constant or a register that holds the value
  - ptr: the register that has the address where the data should be stored



Variable Example

```
 1 #include <stdlib.h>
 2
 3 int g = 0 ;
 4
 5 int main() {
 6  int t = 0;
 7  int * p;
 8  p=malloc(sizeof(int));
 9  free(p);
10 }
```



```
 1  @g = global i32 0, align 4
 …
 8  define i32 @main() #0 {
 …
10  %t = alloca i32, align 4
11 store i32 0, ptr %t, align 4

12 %p = alloca ptr, align 8

13 %call = call noalias ptr
    @malloc(i64 4) #2
14 store ptr %0, ptr %p,
   align 8
15 %1 = load ptr, ptr %p, align 8
 …
```

Integer Conversion (1/2)

- Truncate
  - Syntax: **<res> = trunc <iN1> <value> to <iN2>** where **iN1** and **iN2** are of integer type, and **N1** > **N2**
  - Examples
    - `%X = trunc i32 257 to i8 ;%X becomes i8:1`
    - `%Y = trunc i32 123 to i1 ;%Y becomes i1:true`
    - `%Z = trunc i32 122 to i1 ;%Z becomes i1:false`

Integer Conversion (2/2)

- Zero extension
  - **<res> = zext <iN1> <value> to <iN2>** where **iN1** and **iN2** are of integer type, and **N1** < **N2**
  - Fill the remaining bits with zero
  - Examples
    - `%X = zext i32 257 to i64 ;%X becomes i64:257`
    - `%Y = zext i1 true to i32 ;%Y becomes i32:1`
- Sign extension
  - **<res> = sext <iN1> <value> to <iN2>** where **iN1** and **iN2** are of integer type, and **N1** < **N2**
  - Fill the remaining bits with the sign bit (the highest order bit) of `value`
  - Examples
    - `%X = sext i8 -1 to i16 ;%X becomes i16:65535`
    - `%Y = sext i1 true to i32 ;%Y becomes i32:2`^{32}`-1`

Other Conversions

- Float-to-float
  - `fptrunc .. to`, `fpext .. to`
- Float-to-integer (vice versa)
  - `fptoui .. to`, `fptosi .. to`, `uitofp .. to`, `sitofp .. To`
  - `Ex. %X = fptoui double 123.0 to i32`
- Pointer-to-integer
  - `ptrtoint .. to`, `inttoptr .. to`
- Bitcast
  - `<res> = bitcast <t1> <value> to <t2>`
    where `t1` and `t2` should be different types and have the same size
  - bitcast does not change any bit (i.e., no-op cast)
  - %z = bitcast <2 x int> %v to i64

Computational Instructions

- Binary operations:
  - Add: `add`, `fadd`,`sub`, `fsub`
  - Multiplication: `mul` , `fmul`
  - Division: `udiv` , `sdiv` , `fdiv`
  - Remainder: `urem` , `srem` , `frem`
- Bitwise binary operations
  - shift operations: `shl` , `lshr` , `ashr`
  - logical operations: `and` , `or` , `xor`

Add Instruction

- `<res> = add [nuw][nsw] <iN> <op1>, <op2>`
  - nuw (no unsigned wrap): if unsigned overflow occurs, the result value becomes a **poison value (undefined)**
    - E.g., `add nuw i8 255, i8 1`
  - nsw (no signed wrap): if signed overflow occurs, the result value becomes a poison value
    - E.g., `add nsw i8 127, i8 1`

Control Representation

- The LLVM front-end constructs the control flow graph (CFG) of every function explicitly in LLVM IR
  - A function has a set of basic blocks each of which is a sequence of instructions
  - A function has exactly one entry basic block
  - Every basic block is ended with exactly one *terminator* instruction which explicitly specifies its successor basic blocks if there exist.
    - Terminator instructions: branches (conditional, unconditional), return, etc.

- Due to its simple control flow structure, it is convenient to analyze, transform the target program in LLVM IR

Label, Return, and Unconditional Branch

- A label is located at the start of a basic block
  - Each basic block is addressed as the start label
  - A label `x` is referenced as register `%x` whose type is label
  - The label of the entry block of a function is "`entry`"
- Return **`ret <type> <value> | ret void`**
- Unconditional branch **`br label <dest>`**
  - At the end of a basic block, this instruction makes a transition to the basic block starting with label **`<dest>`**
  - E.g: `br label %entry`

Conditional Branch

- `<res> = icmp <cmp> <ty> <op1>, <op2>`
  - Returns either `true` or `false` (`i1`) based on comparison of two variables (`op1` and `op2`) of the same type (`ty`)
  - `cmp`: comparison option

    `eq` (equal), `ne` (not equal), `ugt` (unsigned greater than), `uge` (unsigned greater or equal), `ult` (unsigned less than), `ule` (unsigned less or equal), `sgt` (signed greater than), `sge` (signed greater or equal), `slt` (signed less than), `sle` (signed less or equal)
- `br i1 <cond>, label <thenbb>, label <elsebb>`
  - Causes the current execution to transfer to the basic block `<thenbb>` if the value of `<cond>` is true; to the basic block `<elsebb>` otherwise.
- Example:

```
1    if (x > y)
2      return 1 ;
3    return 0 ;
```



```
11 %0 = load i32, ptr %x
12 %1 = load i32, ptr %y
13 %cmp = icmp sgt i32 %0, %1
14 br i1 %cmp, label %if.then, label %if.end

15 if.then:
   …
```

Switch

- **`switch <iN> <value>, label <defaultdest> [<iN> <val>, label <dest> …]`**
  - Transfer control flow to one of many possible destinations
  - If the value is found (`val`), control flow is transferred to the corresponding destination (`dest`); or to the default destination (`defaultdest`)
  - Examples:

```
1   switch(x) {
2      case 1:
3         break ;
4      case 2:
5         break ;
6      default:
7        break ;
8   }
```



```
11  %0 = load i32, ptr %x
12  switch i32 %0, label %sw.default [
13    i32 1, label %sw.bb
14    i32 2, label %sw.bb1]

15  sw.bb:
16    br label %sw.epilog

17  sw.bb1:
18    br label %sw.epilog

19  sw.default:
20    br label %sw.epilog

21  sw.epilog:
      …
```

PHI (Φ) instruction

- `<res> = phi <t> [ <val_0>, <label_0>], [ <val_1>, <label_1>], …`
  - Return a value `val_i` of type `t` such that the basic block executed right before the current one is of `label_i`

- Example

```
1  y = (x > 0) ? x : 0 ;
```



```
11 %0 = load i32, ptr %x
12 %c = icmp sgt i32 %0 0
13 br i1 %c, label %c.t, %c.f

14 c.t:
15  %1 = load i32, ptr %x
16  br label %c.end

17 c.f:
18  br label %c.end

19 c.end:
20  %res = phi i32 [%1, %c.t], [0, %c.f]
21  store i32 %res, ptr %y
```

Select

- <result> = `select` <selty> <cond>, <ty> <val1>, <ty> <val2> ;
  - <selty> is either i1 or {<N x i1>}
  - Ex> %X = select i1 true, i8 17, i8 42 ; yields i8:17

- The ‘select‘ instruction is used to choose one value based on a condition, without IR-level branching.
  - If <cond> ==1, the instruction returns the first value argument; the second value argument otherwise
  - If the condition is a vector of i1, then the value arguments must be vectors of the same size, and the selection is done element by element.
  - If the condition is an i1 and the value arguments are vectors of the same size, then an entire vector is selected.

Function Call

- **<res> = call <ResType> [<FunType>] @<FunName>([arg list])**
  - <ResType>: the type of the call return value
  - <FunType>: the signature of the pointer to the target function (optional)
  - @<FunName>: an LLVM value containing a pointer to a target function
  - [arg list]: argument list whose types match the function signature

- Examples:

```
1  printf("%d", abs(x));
```



```
11 @.str = constant [3 x i8] c"%d\00"

12 %0 = load i32, ptr %x
13 %abs_x  = call i32 @abs(i32 %0)

14 %1 = call i32 (ptr, ...)
     @printf(ptr @.str, i32 %abs_x)
```

Function Declaration

- **`declare <ResType> @<FunName>([arg list])`**
  - `<ResType>`: the type of the call return value
  - `@<FunName>`: an LLVM value containing a pointer to a target function
  - `[arg list]`: argument list whose types match the function signature
- Examples:

```
int printf(char *,...)
```



```
declare i32 @printf(ptr, ...)
```

A Simple Example

```
 1 int sum(int x) {
 2   int i, sum=0;
 3   for (i=1; i <=x; i++)
 4     sum += i;
 5   return sum;
 6 }
 7
 8 int main() {
 9   printf("%d\n",sum(4));
10 }
```



```
 1 declare i32 @printf(ptr, ...)
 2
 3 @.str = global [4 x i8] c"%d\0A\00"
 4
 5 define i32 @sum(i32 %x) {
 6 entry:
 7   %x.addr = alloca i32
 8   %i = alloca i32
 9   %sum = alloca i32
10   store i32 %x, ptr %x.addr
11   store i32 0, ptr %sum
12   store i32 1, ptr %i
13   br label %for.cond
14
15 for.cond:  ; preds = %for.inc, %entry
16   %0 = load i32, ptr %i
17   %1 = load i32, ptr %x.addr
18   %cmp = icmp sle i32 %0, %1
19   br i1 %cmp, label %for.body, label %for.end
20
21 for.body:   ; preds = %for.cond
22   %2 = load i32, ptr %i
23   %3 = load i32, ptr %sum
24   %add = add i32 %3, %2
25   store i32 %add, ptr %sum
26   br label %for.inc
27
28 for.inc:   ; preds = %for.body
29   %4 = load i32, ptr %i
30   %inc = add i32 %4, 1
31   store i32 %inc, ptr %i
32   br label %for.cond
33
34 for.end:   ; preds = %for.cond
35   %5 = load i32, ptr %sum
36   ret i32 %5
37 }
38
39 define i32 @main() {
40   %sum = call i32 @sum(i32 4)
41   %printf= call i32 (ptr, ...) @printf(@.str,
i32 %sum)
42   ret i32 0
43 }
```

Aggregate Types and Function Type

- Array: **[<# of elements> x <type>]**
  - Single dimensional array ex: `[40 x i32],[4 x i8]`
  - Multi dimensional array ex: `[3 x [4 x i8]],[12 x [10 x float]]`

- Structure: **type {<a list of types>}**
  - E.g. `type{ i32, i32, i32 },type{ i8, i32 }`

- Function: **<return type> (a list of parameter types)**
  - E.g. `i32 (i32), float (i16, ptr)`

`Getelementptr` Instruction

- A memory in an aggregate type variable can be accessed by **`load/store`** instruction and **`getelementptr`** instruction that obtains the pointer to the element.

- Syntax:

```
<res> = getelementptr <pty>, ptr <ptrval>{,<t> <idx>}*
```

  - res: the target register
  - pty: the register that defines the aggregate type
  - ptrval: the register that points to the data variable
  - t: the type of index
  - idx: the index value

Aggregate Type Example 1 (1/2)

```
1   struct pair {
2     int first;
3     int second;
4   };

5   int main() {
6     int arr[10];
7     struct pair a;

8     a.first = arr[1];
      …
```

```
11 %struct.pair = type{ i32, i32 }

12 define i32 @main() {
13  entry:
14    %arr = alloca [10 x i32]
15    %a = alloca %struct.pair

16    %arrayidx = getelementptr
        [10 x i32], ptr %arr, i64 0,
    i64 1

17    %0 = load i32, ptr %arrayidx

18    %first = getelementptr
       %struct.pair, ptr %a, i32 0,
    i32 0

19    store i32 %0, ptr %first
```

Aggregate Type Example 1 (2/2)

```
%first=getelementptr %struct.pair, ptr %a,i32 0,i32 0
```

1. The first operand of `getelementptr` (e.g., `%a`) is a pointer to a data structure.
2. An element of an aggregate data structure must be accessed by `getelementptr` with a pointer (e.g., `%a`) and an offset index (`a.first== (&a)[0].first`, and/or `(&a)->first == (&a)[0].first`)

```
struct point {
  int x;
  struct point* p;};

int main() {
  struct point s, s1;
  s.p=&s1; // (&s)[0].p=&s1;
  (s.p)->x = 0;
}
```

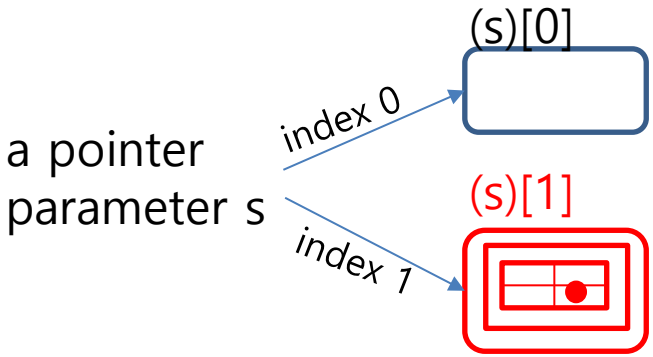
```
%struct.point = type { i32, ptr }
define i32 @main() #0 {
entry:
  %s = alloca %struct.point, align 8
  %s1 = alloca %struct.point, align 8
  // s.p=&s1;
  %p = getelementptr  %struct.point, ptr %s, i32 0, i32 1
  store ptr %s1, ptr %p, align 8
  // (s.p)->x = 0
  %p1 = getelementptr  %struct.point, ptr %s, i32 0, i32 1
  %0 = load ptr, ptr %p1, align 8
  %x = getelementptr  %struct.point, ptr %0, i32 0, i32 0
  store i32 0, ptr %x, align 4

  ret i32 0
}
```

Aggregate Type Example 2

```
 1 struct RT {
 2   char A;
 3   int B[10][20];
 4   char C;
 5 };
 6 struct ST {
 7   int X;
 8   double Y;
 9   struct RT Z;
10 };

11 int *foo(struct ST *s) {
12   return &(s[1].Z.B[5][13]);
13 }
```




```
1 %struct.RT = type { i8,
      [10 x [20 x i32]], i8 }
2 %struct.ST = type { i32,
      double, %struct.RT }

3 define ptr @foo(ptr %s) {
  entry:
4   %0 = getelementptr // s[1]
      %struct.ST, ptr %s, i64 1
5   %1 = getelementptr // s[1].Z
      %struct.ST, ptr %0, i32 0, i32 2
6   %2 = getelementptr // s[1].Z.B
      %struct.RT, ptr %1, i32 0, i32 1
7   %3 = getelementptr //s[1].Z.B[5][13]
      [10 x [20 x i32]], ptr %2, i64 0,
                         i64 5, i64 13
8   ret ptr %3
9 }
```

Aggregate Type Example 3

```
%MyVar = global { [10 x i32] }
%idx1=getelementptr {[10xi32]}, ptr %MyVar,i64 0,i32 0,i64 1
%idx2=getelementptr {[10xi32]}, ptr %MyVar,i64 1
```

- `idx1` computes the address of the 2nd integer in the array that is in the structure in %MyVar, that is MyVar+4.
  - The type of `idx1` is i32*.
- `idx2` computes the address of the next structure after %MyVar. The value of `idx2` is MyVar + 40 because it indexes past the ten 4-byte integers in MyVar.
  - The type of `idx2` is { [10 x i32] }*

Aggregate Type Example 4

```
%MyVar = global { [10 x i32] }
%idx1=getelementptr {[10xi32]}, ptr %MyVar,i64 1,i32 0,i64 0
%idx2=getelementptr {[10xi32]}, ptr %MyVar,i64 1
```

- The value of `%idx1` is `%MyVar`+40
  - its type is i32*
- The value of `%idx2` is also `%MyVar`+40
  - but its type is { [10 x i32] }*.

Unaddressed Issues

- Many options/attributes of instructions
- Vector data type (SIMD style)
- Exception handling
- Object-oriented programming specific features
- Concurrency issues
  - Memory model, synchronization, atomic instructions

* *http://llvm.org/docs/LangRef.html*